

LE_BMS_A_V2 Fabrication Document

Layer Stack Legend

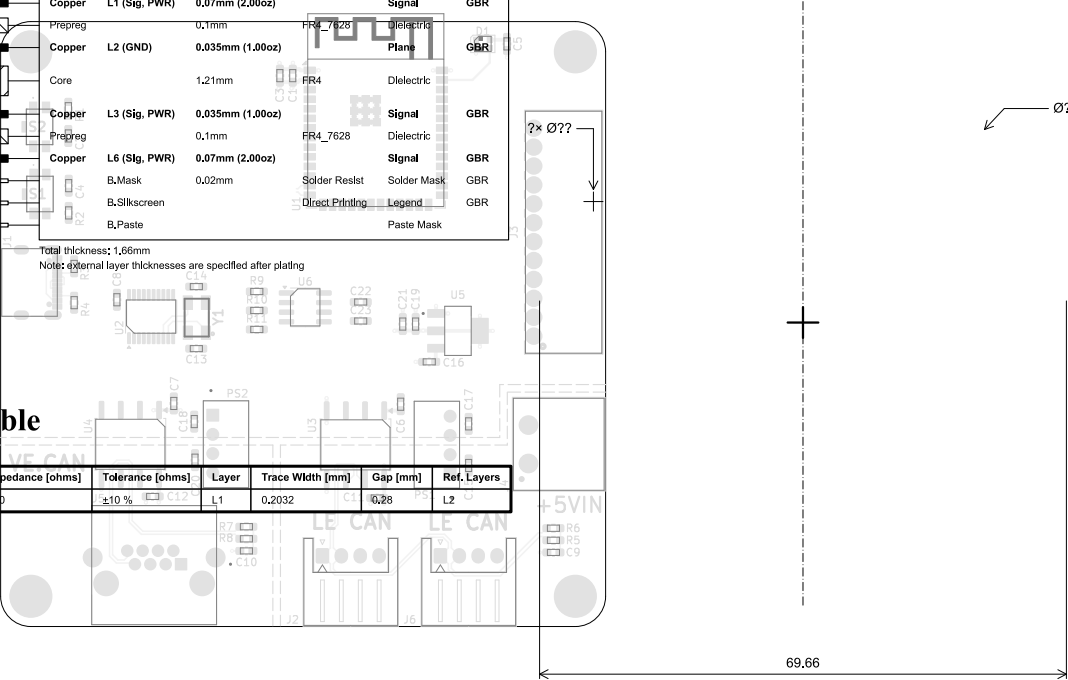
Material	Layer	Thickness	Dielectric	Type	Gerber
F,Paste				Paste Mask	
F,Silkscreen				Legend	GBR
F,Mask		0,02mm		Solder Mask	GBR
Copper	L1 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
Prepreg		0,1mm	FR4_7628	Dielectric	
Copper	L2 (GND)	0,035mm (1,00oz)		Plane	GBR
Core		1,21mm	FR4	Dielectric	
Copper	L3 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
Prepreg		0,1mm	FR4_7628	Dielectric	
Copper	L6 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
B,Mask		0,02mm		Solder Mask	GBR
B,Silkscreen				Legend	GBR
B,Paste				Paste Mask	

Total thickness: 1,66mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0,2032	0,28	L2

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

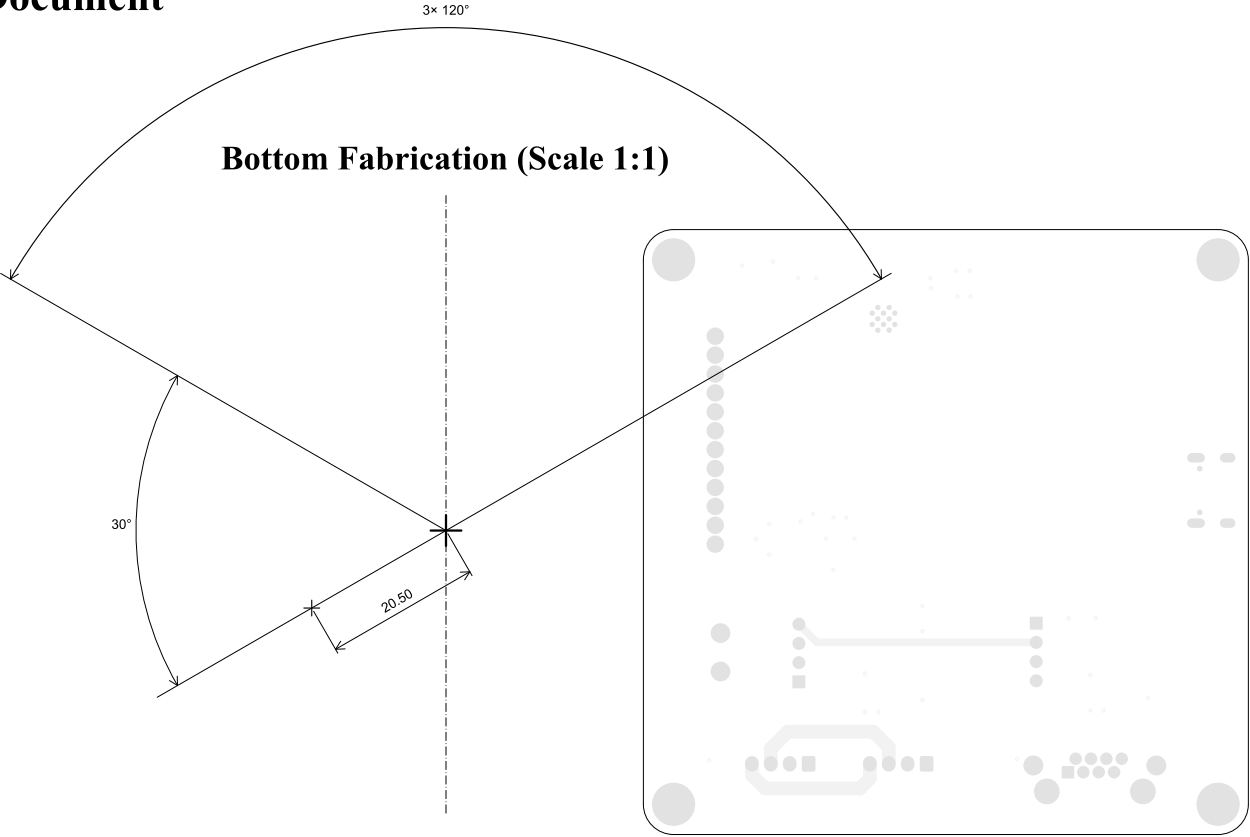
SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:
 - A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
 - B. Tg 170 C OR EQUIVALENT.
 - C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY COMPANY.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	80,000 × 80,000 mm
BOARD THICKNESS	1,660 mm
TRACE WIDTH	0,200 mm
TRACE TO TRACE	0,200 mm
MIN. HOLE (PTH)	0,200 mm
MIN. HOLE (NPTH)	0,650 mm
ANNULAR RING	0,150 mm
COPPER TO HOLE	0,254 mm
COPPER TO EDGE	0,250 mm
HOLE TO HOLE	0,254 mm
- 11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.
- 12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

	Comments:	Company: Company Name		Variant: PRELIMINARY	Git Hash: 5f6d398
	Sheet Title: Top Fabrication (Scale 1:1)	Board Name: LE_BMS_A_V2		Project Name: LE_V2	
	Sheet Path:	File Name: LE_BMS_A_V2.kicad_pcb	Designer: Author	Date: 2024-04-13	Revision: + (Unreleased)
			Reviewer:	Size: A4	Sheet: 1 of 10

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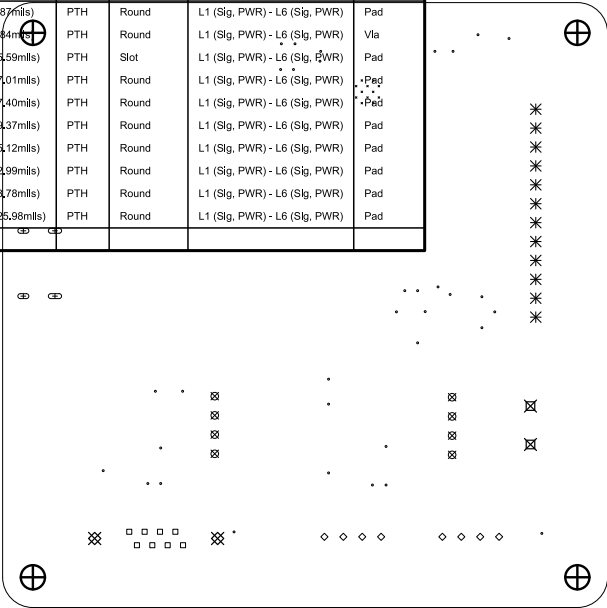
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	Bottom Fabrication (Scale 1:1)	LE_BMS_A_V2.kicad_pcb	Author	2024-04-13	+ (Unreleased)
	Sheet Path:		Reviewer:	Size:	Sheet:
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	12	0,20mm (7,87mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
○	34	0,25mm (9,84mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Via
+	4	0,65mm (25,59mils)	PTH	Slot	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
□	8	0,94mm (37,01mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
◇	8	0,95mm (37,40mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊠	8	1,00mm (39,37mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⌘	12	1,40mm (55,12mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊞	2	1,60mm (62,99mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊞	2	1,62mm (63,78mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
⊕	4	3,20mm (125,98mils)	PTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Pad
Total 94						

Drill Drawing L1 - L4 (Scale 1:1)



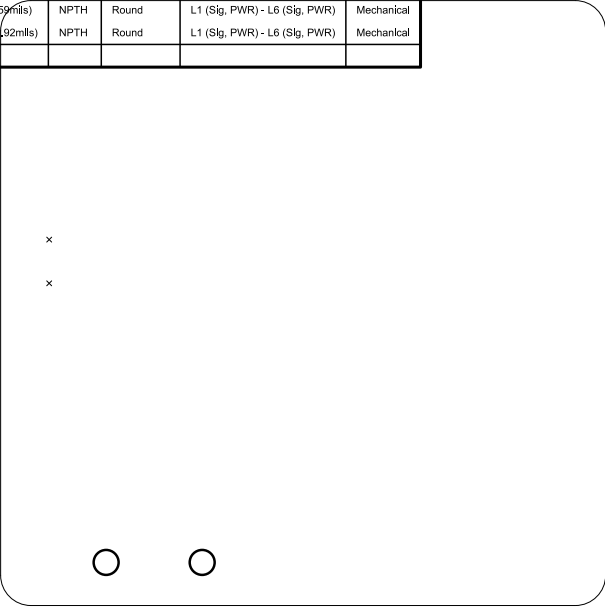
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		Board Name: LE_BMS_A_V2		Project Name: LE_V2	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: LE_BMS_A_V2.kicad_pcb	Designer: Author	Date: 2024-04-13	Revision: + (Unreleased)
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	2	0,85mm (25,59mils)	NPTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Mechanical
O	2	3,30mm (125,92mils)	NPTH	Round	L1 (Sig, PWR) - L6 (Sig, PWR)	Mechanical
	Total 4					

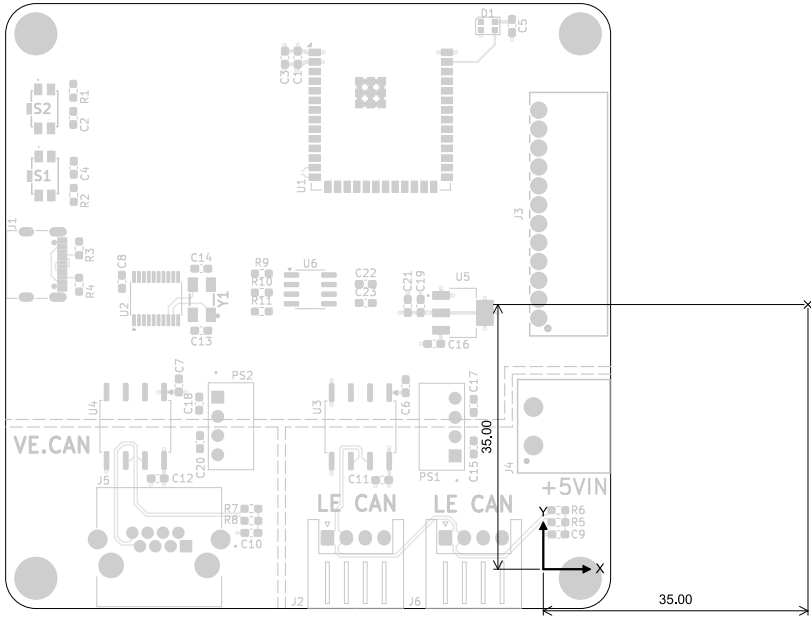
Drill Drawing L1 - L4 (Scale 1:1)



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Top Test Points (Scale 1:1)

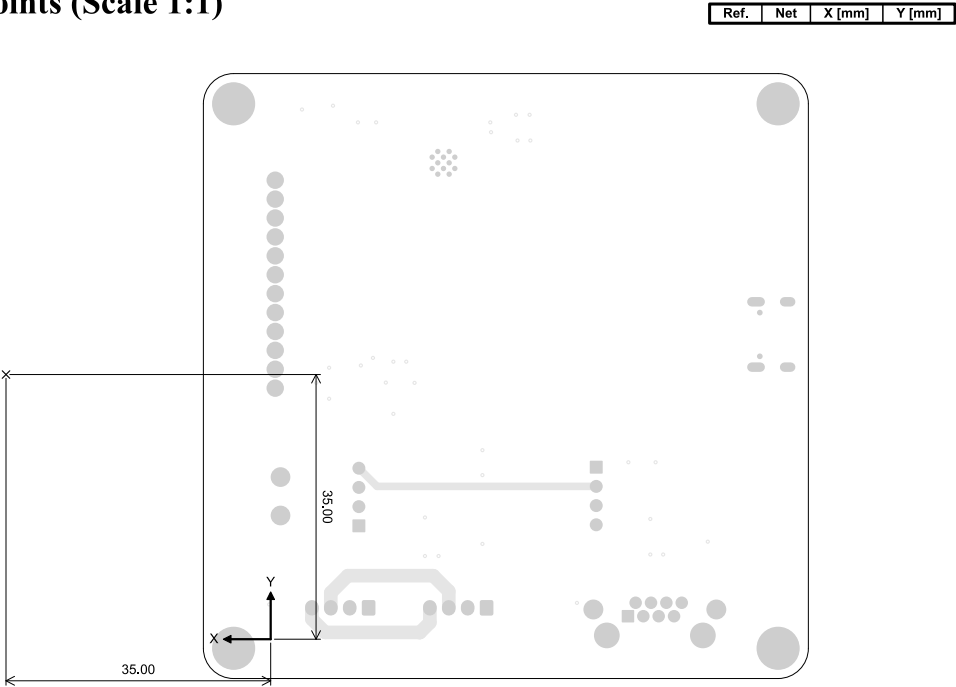


All dimensions are in millimeters unless otherwise specified.

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Bottom Test Points (Scale 1:1)

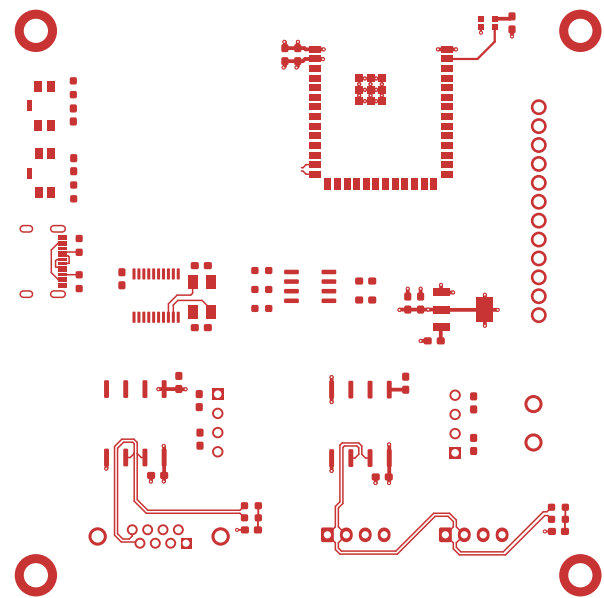


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Bottom Test Points (Scale 1:1)	LE_BMS_A_V2.kicad_pcb		Author	2024-04-13	+ (Unreleased)		
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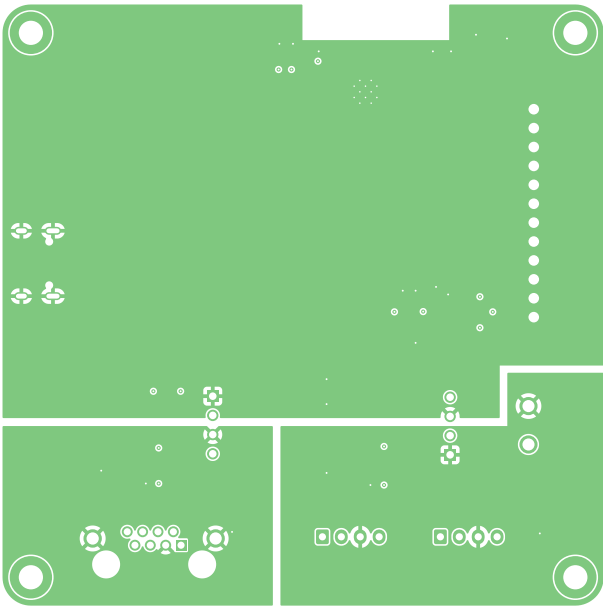
L1 (Sig, PWR) (Scale 1:1)



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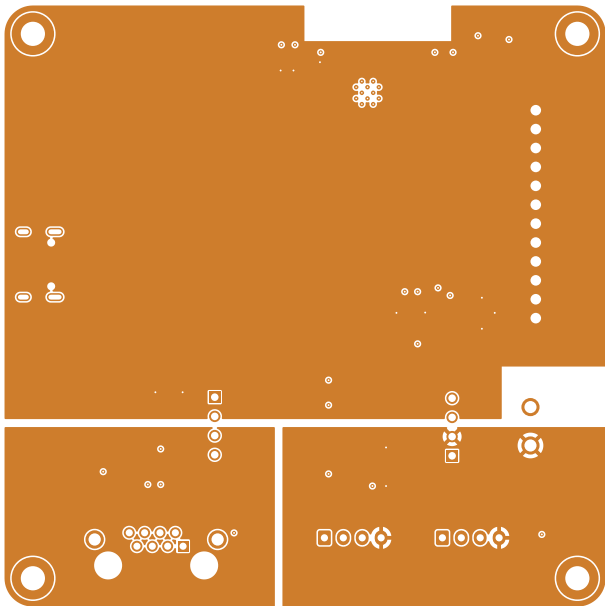
L2 (GND) (Scale 1:1)



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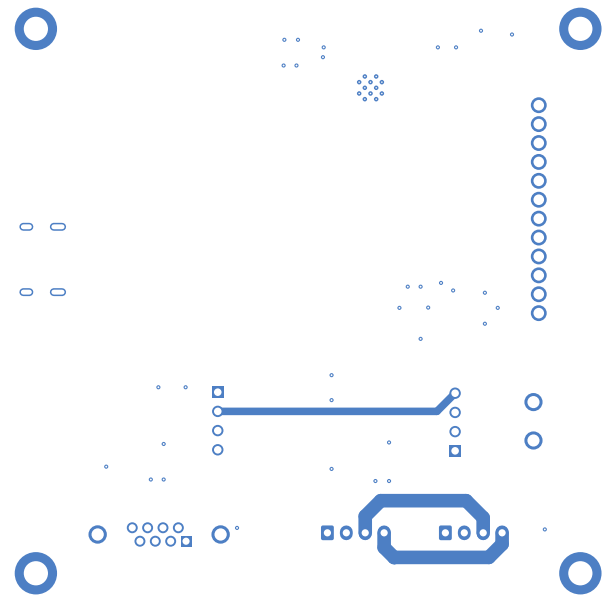
L3 (Sig, PWR) (Scale 1:1)



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L6 (Sig, PWR) (Scale 1:1)



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